DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

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TUTORIAL 1: LITERATURE REVIEW AND RESEARCH GAP

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Project Title:	FinReg-SLM: Domain-Specialized Small Language Model for Indian Financial Compliance
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PART B — Project Title & Background

1. Project Title

FinReg-SLM: A Version-Aware, Multi-Agency Retrieval-Augmented Fine-Tuned Small Language Model with Native Chain-of-Thought Reasoning and Direct Preference Optimization for Indian Financial Regulatory Compliance

2. Background & Motivation

Regulatory Environment in India: Indian financial institutions operate under one of the strictest regulatory environments in the world. Thirteen statutory bodies—such as the Reserve Bank of India (RBI), the Securities and Exchange Board of India (SEBI), the Ministry of Corporate Affairs (MCA), and the Central Board of Indirect Taxes and Customs (CBIC)—regularly publish circulars, notifications, Master Directions, and statutory guidelines. Together with 25 governing Parliamentary Acts, there are over 18,242 active regulatory documents. Commercial banks, Non-Banking Financial Companies (NBFCs), asset management companies, and fintech startups must adhere to all of these rules simultaneously. A single compliance failure can lead to heavy financial penalties, interest charges, and operational license cancellations.

Temporal Volatility and Superseded Rules: A key practical challenge in compliance is that regulations change frequently. Amendments routinely modify, supersede, or clarify earlier clauses. For example, a 2025 SEBI amendment to the Takeover Regulations changed share valuation requirements for open offers while providing a 9-month transition period for ongoing assignments. When organizations use standard semantic search or basic Retrieval-Augmented Generation (RAG), the system retrieves older circulars simply because they match the query's keywords, presenting outdated provisions as active law.

Data Privacy and the DPDP Act 2023: Most commercial AI compliance tools rely on cloud-hosted frontier models like GPT-4o or Claude 3.5. However, uploading internal financial audit logs, customer records, and confidential banking data to third-party offshore APIs violates India's Digital Personal Data Protection (DPDP) Act 2023. Financial institutions require self-hosted, on-premises models that keep all data within their own secure infrastructure.

Hallucination and Epistemic Limits: General-purpose language models also exhibit specific failure modes in statutory question answering: (1) Hallucination of Authority, where models invent plausible-sounding but fictitious circular numbers; (2) Distractor Sensitivity, where irrelevant context passages confuse the model; and (3) Sycophancy, where models answer questions based on false premises rather than pointing out the statutory error.

The FinReg-SLM Approach: FinReg-SLM addresses these issues by adapting a 7B Small Language Model (DeepSeek-R1-Distill-Qwen-7B) for local execution on standard hardware (8GB VRAM). The architecture combines a version-aware hybrid retrieval engine (BM25 lexical search + dense vector embeddings + Reciprocal Rank Fusion + date filtering) with a Two-Stage Alignment pipeline (Stage 1 QLoRA fine-tuning on step-by-step reasoning traces + Stage 2 Direct Preference Optimization for citation grounding and epistemic refusal).

3. Problem Statement

Indian financial institutions governed by 13 regulatory bodies lack automated, version-aware compliance verification tools. Existing general-purpose language models exhibit a 17%–33% hallucination rate, fail to track statutory amendments, confabulate answers on out-of-scope queries, and violate data privacy mandates under the DPDP Act 2023. This project designs, implements, and evaluates an on-premises Small Language Model

architecture that delivers verifiable citation grounding, calibrated refusal, and version-aware multi-agency reasoning on consumer-grade hardware.

PART C — Literature Search Methodology

4. Database Search Strategy

A systematic search was carried out across Scopus, ScienceDirect (Elsevier), IEEE Xplore, and ACM Digital Library. The search targeted three core areas: (1) Indian financial and regulatory NLP, (2) Retrieval-Augmented Fine-Tuning (RAFT) and Small Language Models (SLMs), and (3) Alignment methodologies for hallucination reduction, catastrophic forgetting mitigation, and temporal validity tracking.

5. Keywords & Search Query

Query: ('Retrieval-Augmented Generation' OR 'RAFT') AND ('Fine-Tuning' OR 'LoRA' OR 'DPO') AND ('Legal NLP' OR 'Financial Compliance' OR 'Regulatory QA') AND ('Hallucination' OR 'Catastrophic Forgetting' OR 'Temporal Validity').

6. Selection Criteria

• Inclusion: Peer-reviewed journal articles and core conference proceedings (EMNLP, NeurIPS, ICLR, COLM, IEEE Access, Elsevier Q1); studies evaluating SLMs (1B–8B) or preference alignment (DPO); papers with reproducible quantitative metrics.
• Exclusion: Non-academic blog posts, generic non-domain conversational chatbots, and papers lacking empirical evaluation.

PART D — Core Literature Review (Top 8 Papers)

7. Table 1: Index of the 8 Primary Literature Sources (with Indexing & DOI)

Sr.	Authors & Year	Paper Title	Venue / Journal	Year	Indexing (Scopus / WoS / IEEE / Core)	DOI / Publication Link
1	Ovadia et al. (2024)	Fine-Tuning or Retrieval? Comparing Knowledge Injection in LLMs	EMNLP 2024 (Main)	2024	Scopus / WoS / ACL Anthology	doi: 10.18653/v1/2024.emnlp-main.15
2	Zhang et al. (2024)	RAFT: Adapting Language Model to Domain Specific RAG	COLM 2024	2024	Peer-Reviewed Core AI / OpenReview	https://openreview.net/forum?id=rzQGHXNReU
3	Magesh et al. (2025)	Hallucination-Free? Assessing Reliability of Leading AI Legal Research Tools	J. Empir. Leg. Stud. (vol. 22)	2025	Scopus Q1 / WoS / Wiley	doi: 10.1111/jels.12413
4	Pai & Hsu (2026)	Fine-Tuning SLMs for Industrial RAG: Factuality & Hallucination	Comput. Stand. Interfaces (vol. 98)	2026	Scopus Q1 / WoS / Elsevier	doi: 10.1016/j.csi.2026.104163
5	Kim (2026)	Detecting Jurisdiction Compliance Gaps with Regulatory RAG	J. Loss Prev. Process Ind. (vol. 103)	2026	Scopus Q1 / WoS / Elsevier	doi: 10.1016/j.jlp.2026.106100
6	Rafailov et al. (2023)	Direct Preference Optimization: Language Model as a Reward Model	NeurIPS 2023 (vol. 36)	2023	Scopus / WoS / NeurIPS Core	https://proceedings.neurips.cc/paper/2023/hash/a85b405265de2473f47cfda014f740f0-Abstract-Conference.html
7	Biderman (2024) / Luo (2023)	LoRA Learns Less & Forgets Less / Catastrophic Forgetting in LLMs	TMLR 2024 / EMNLP 2023	2023-24	Scopus / WoS / ACL & TMLR	doi: 10.18653/v1/2023.emnlp-main.555
8	Rajveer Singh Pall (2026)	IndiaFinBench: Benchmark for LLMs on Indian Financial Regulatory Text	arXiv:2604.19298	2026	Standardized Benchmark / Hugging Face	doi: 10.48550/arXiv.2604.19298 (hf: Rajveer-code/IndiaFinBench)

8. Table 2: Comparative Literature Review Matrix (Problem, Gap, Fix & Impact)

Sr.	Authors & Year	Research Problem	Dataset	Method	Metrics	Key Findings	Identified Gap	Our Applied Fix	Empirical Score Gain
1	Ovadia et al. (2024)	RAG vs Fine-Tuning for knowledge injection	Synthetic facts & QA	Empirical comparison of RAG vs SFT	Accuracy, Reversal Rate	RAG beats SFT; fine-tuned models suffer from the Reversal Curse	Proves FT alone fails for changing facts, but misses hybrid CoT reasoning	Two-Stage RAFT with <think> CoT traces	**Core compliance: 62.3% -> 94.43%; IndiaFinBench Reg: 90.29%**
2	Zhang et al. (2024)	Model confusion from distractor passages in RAG	PubMed, HotpotQA	RAFT (Golden + Distractor SFT)	Top-1 Extraction Accuracy	Training with distractors forces model to quote evidence verbatim	Tested on general QA; not tested on legal circular amendments	Extended RAFT to Indian law (257 CoT traces with distractors)	**Multi-hop synthesis (C2) improved from 54.0% to 94.44% (+40.44%)**
3	Magesh et al. (2025)	Hallucination in commercial legal AI tools	200+ legal queries	Pre-registered empirical audit	Hallucination Rate, Citation Grounding	17%-33% hallucination rate; frequent phantom citations and sycophancy	Audited closed commercial tools without an open-weights fix	Verbatim citation verification + DPO margin loss penalty	**Citation Grounding: 51.6% -> 99.17%; Phantom Citations: 0.0%**
4	Pai & Hsu (2026)	Small LLM hallucination on internal regulations	Industrial RAG logs	QLoRA on 1.5B-8B SLMs	5-category hallucination taxonomy	Regulatory RAG is harder than customer service (0.488 vs 0.661 mean)	Corporate guidelines focus; lacked legislative amendment tracking	4-bit QLoRA on DeepSeek-R1-7B on 6 failure categories (C1-C6)	**Cross-regulatory overlap: 96.67%; 100.0% structural conformance**
5	Kim (2026)	Jurisdiction-specific compliance checking	1,327 Korean safety rules	SQLite FTS5 + Qwen-4B (Ollama)	Compliance Detection Rate	On-premises SLMs raised local violation detection from 0/60 to 26/60	Single-agency focus; lacked temporal amendment tracking	Multi-agency SQLite WAL index + effective_date temporal filter	**IndiaFinBench Contradiction: 79.00%; Temporal: 75.78%**
6	Rafailov et al. (2023)	Reinforcement learning instability in alignment	Anthropic HH-RLHF	Direct Preference Optimization (DPO)	Win Rate vs GPT-4	DPO optimizes policy directly from pairs without reward models	General helpfulness focus; lacked legal citation grounding and refusal	257 domain DPO pairs with hallucination traps & refusal targets	**Epistemic Refusal: 0.0% -> 82.50%; Premise Rejection: 0.0% -> 58.00%**
7	Biderman / Luo (2023-24)	Catastrophic forgetting under PEFT fine-tuning	1B-7B LLM benchmarks	Empirical forgetting audit (LoRA)	Accuracy drop across tasks	Full FT degrades general reasoning; LoRA preserves base weights	Did not optimize adapter rank across all 7 projection matrices	All-Linear QLoRA (Rank 64, Alpha 128) in 4-bit NF4 + ChatML	**100.0% Structural Protocol Conformance; zero formatting loss**
8	Rajveer Singh Pall (2026)	Benchmarking LLMs on Indian financial regulations	192 SEBI & RBI documents	Zero-shot benchmark evaluation	Accuracy across 4 task slices	Base 7B/8B models score ~76%; numerical reasoning is hardest (35.9% spread)	Released benchmark only; did not build a fine-tuned SLM	Evaluated Stage 2 Aligned 7B model on official 324-question test split	**Overall Accuracy: 80.98% (beats Base 7B/8B models by +3.4% to +5.0%)**

PART E — Literature Synthesis

9. Synthesis Across Key Themes

Theme 1: SLMs vs. Cloud Models: The literature shows a clear shift toward domain-specialized Small Language Models (3B–8B parameters). By combining Low-Rank Adaptation (Hu et al. [7]) and Direct Preference Optimization (Rafailov et al. [6]), small models can achieve high accuracy on specialized tasks while running locally, avoiding cloud API costs.

Theme 2: Hybrid Retrieval and Version Awareness: Standard semantic vector search struggles with exact alphanumeric circular numbers and superseded clauses. Combining dense semantic embeddings with BM25 keyword matching via Reciprocal Rank Fusion ($k=60$) and adding an explicit temporal date filter (`as_of_date` vs `effective_date`) provides a lightweight, effective solution (Kim [5], Cormack [5]).

Theme 3: Epistemic Abstention and Sycophancy: Standard instruction fine-tuning teaches models to always generate an answer, even when given false premises or out-of-scope questions (Magesh et al. [3], Pai & Hsu [4]). In regulatory compliance, hallucinating an answer is far more dangerous than stating that information is missing. Preference optimization with negative pairs trains the model to actively refuse when statutory context is absent.

Theme 4: Data Sovereignty and On-Premises Deployment: Under India's DPDP Act 2023, financial institutions face strict restrictions on sharing sensitive audit logs with third-party cloud APIs. Deploying a 4-bit quantized 7B model locally on standard hardware (8GB VRAM) guarantees complete data privacy and eliminates recurring per-token API costs.

PART F — Research Gaps, Applied Fixes & Empirical Gains

10. Identified Research Gaps to Solutions Mapping

Gap 1 (RAG vs FT Dichotomy): Literature establishes that RAG provides dynamic facts while fine-tuning sets tone and style. However, naive RAG gets confused by legal distractors. Fix: Two-Stage RAFT with native <think> reasoning traces on golden + distractor circulars. Result: Core statutory compliance rose from 62.3% to 94.43% on G7; IndiaFinBench Regulatory Interpretation scored 90.29%.

Gap 2 (Hallucination of Authority): Legal AI tools have a 17%–33% hallucination rate, frequently fabricating circular numbers. Fix: Verbatim citation verification and DPO margin loss penalizing ungrounded references. Result: Citation Grounding increased from 51.6% to 99.17%; phantom citation rate dropped to 0.0%.

Gap 3 (Temporal Amnesia): Semantic search retrieves obsolete circulars when regulations are amended. Fix: Structured date metadata indexing (`effective_date`) + BM25/Dense RRF + temporal DPO training. Result: Contradiction Detection scored 79.00%; Temporal Reasoning scored 75.78% on IndiaFinBench.

Gap 4 (Epistemic Inability to Abstain): Base models answer ungrounded queries and accept false premises due to sycophancy. Fix: Negative preference DPO alignment on out-of-scope and false-premise pairs. Result: Epistemic Refusal jumped from 0.0% to 82.50%; False Premise Rejection jumped from 0.0% to 58.00%.

Gap 5 (Catastrophic Forgetting): Fine-tuning on legal text often degrades formatting and reasoning abilities. Fix: All-linear QLoRA (Rank 64, Alpha 128) across all 7 projection matrices in 4-bit NF4 with strict ChatML supervision. Result: 100.0% Structural Protocol Conformance with zero formatting collapse.

Gap 6 (Non-Western Regulatory Evaluation Void): Existing financial NLP benchmarks focus solely on US SEC/GAAP filings. Fix: Comprehensive evaluation on both our internal G7 300-query suite and the independent IndiaFinBench

324-question test split using an automated 70B LLM judge. Result: IndiaFinBench Overall Macro Accuracy of 80.98%, outperforming Base 7B/8B open LLMs by +3.4% to +5.0%.

PART G & H — Research Question & Objectives

11. Core Research Question

How does combining a version-aware multi-agency hybrid retrieval pipeline (Dense Vector + BM25 Lexical + Reciprocal Rank Fusion + Temporal Filtering) with a Two-Stage Aligned Reasoning Model (Stage 1 QLoRA SFT on Chain-of-Thought traces + Stage 2 Native DPO on epistemic preference pairs using DeepSeek-R1-Distill-Qwen-7B) compare against baseline small language models and single-stage SFT in achieving factual accuracy, 100% citation grounding, and calibrated epistemic abstention across the 13 Indian financial regulatory authorities under on-premises DPDP Act 2023 constraints?

12. Research Objectives

1. Index an exhaustive corpus of 18,242 Indian financial regulatory documents (435 MB SQLite WAL) across 13 authorities and 25 Parliamentary Acts with SHA-256 deduplication.

2. Implement a version-aware hybrid retrieval engine (Dense BGE-small + BM25 + Reciprocal Rank Fusion k=60 + Temporal Filter on effective_date).

3. Synthesize 257 golden reasoning SFT traces and 257 DPO preference pairs containing hallucination and sycophancy traps.

4. Train DeepSeek-R1-Distill-Qwen-7B via Stage 1 QLoRA SFT and Stage 2 Fused-Kernel DPO on a single consumer 8GB GPU.

5. Merge adapters into standalone 15.2 GB FP16 SafeTensors weights for private, on-premises DPDP Act 2023 compliant deployment.

6. Evaluate the system across the 300-query G7 Suite and the 324-question IndiaFinBench test split using an automated Frontier LLM Judge (Meta-Llama-3.3-70B-Instruct).

PART I & J — Scope & Empirical Benchmark Results

13. Scope of the Investigation

Data Scope: 18,242 verified documents & 25 Acts across 13 Indian regulators (RBI, SEBI, MCA, CBIC, CBDT, etc.).

Population Scope: Scheduled commercial banks, NBFCs, fintechs, and compliance auditing professionals in India.

Technical Scope: Small Reasoning Models (7B) adapted via 4-bit QLoRA and Native DPO for local execution.

Evaluation Scope: Dual benchmark: 300-query G7 Suite + official 324-question IndiaFinBench test split.

Application Scope: On-premises compliance verification and audit support under the DPDP Act 2023.

14. Empirical Benchmark Results (G7 Suite & IndiaFinBench)

Table A: Internal G7 Hard Compliance Benchmark (300 Queries — Evaluated via LLaMA-3.3-70B Judge):

Model Configuration	C1 (Direct)	C2 (Multi-Hop)	C3 (Cross-Reg)	C4 (Numeric)	C5 (Abstain)	C6 (Reject)	Citation Rate	LLM Judge Score
Base 3B (Llama-3.2)	58.0%	52.0%	46.0%	44.0%	0.0% (Hallucinated)	0.0% (Sycophantic)	18.6%	44.80%
RAFT Fine-Tuned 3B	72.0%	68.0%	64.0%	60.0%	0.0% (Hallucinated)	0.0% (Sycophantic)	78.0%	18.80%*
Stage 1 SFT 7B (R1)	90.0%	92.0%	89.5%	91.0%	0.0% (Hallucinated)	0.0% (Sycophantic)	88.4%	68.17%
Stage 2 DPO 7B (Ours)	**92.2%**	**94.4%**	**96.7%**	**93.8%***	**82.5% (Refusal)**	**58.0% (Rejection)**	**99.2%**	**70.71%**

Table B: Complete IndiaFinBench Test Split Leaderboard (arXiv:2604.19298 — 324 Questions across SEBI & RBI):

Model Name	Model Size & Hosting	Regulatory Interp.	Contradiction	Temporal	Numerical	Overall Accuracy	Benchmark Standing
Gemini 2.5 Flash	Cloud MoE	93.40%	88.00%	86.50%	87.20%	89.70%	Tier 1 (Proprietary Cloud API)
GPT-4o	Cloud MoE	92.10%	86.00%	84.40%	85.10%	88.20%	Tier 1 (Proprietary Cloud API)
Claude 3.5 Sonnet	Cloud MoE	91.00%	84.00%	82.80%	83.50%	86.50%	Tier 1 (Proprietary Cloud API)
FinReg-SLM 7B (Ours)	**7B (Local 8GB)**	**90.29%**	**79.00%**	**75.78%**	**69.18%**	**80.98%**	**Top Tier 2 (Beats All Base Models)**
LLaMA-3-8B Base	8B (General)	84.20%	72.50%	70.10%	58.40%	77.60%	Tier 2 (Open Base Model)
Mistral-7B-Instruct	7B (General)	82.80%	71.00%	68.50%	56.20%	76.20%	Tier 2 (Open Base Model)
Gemma 4 E4B	4B (Compact)	76.50%	65.00%	62.00%	51.00%	70.40%	Tier 3 (Compact Open Model)
Non-Specialist Human	Human Baseline	74.00%	66.00%	65.00%	54.00%	69.00%	Human Non-Specialist Baseline

Benchmark Performance Analysis: Comparative Analysis of Benchmark Tiers and Trade-offs:

1. Frontier Cloud Models (Tier 1: Gemini 2.5 Flash, GPT-4o, Claude 3.5 Sonnet): These proprietary multi-hundred-billion parameter models achieve 86.50%–89.70% overall accuracy due to massive multi-trillion token pretraining and extensive cloud compute clusters. However, sending confidential banking audit logs and internal financial communications to third-party offshore APIs violates data sovereignty covenants under India's DPDP Act 2023.

Furthermore, their recurring per-token API costs are prohibitive for continuous enterprise compliance monitoring.
2. FinReg-SLM 7B Performance (Ours): Running entirely on an on-premises consumer 8GB GPU with zero cloud data transfer, FinReg-SLM achieves 80.98% overall macro accuracy. Crucially, on Regulatory Interpretation (139 statutory questions), our model scores 90.29%, directly rivaling GPT-4o (92.10%) and Claude 3.5 Sonnet (91.00%). Furthermore, FinReg-SLM outperforms all general open-source base models of comparable size (LLaMA-3-8B at 77.60% and Mistral-7B at 76.20%) by +3.4% to +5.0%, and exceeds the non-specialist human baseline (69.00%) by +11.98%.
3. The Numerical Reasoning Gap (69.18%): Multi-step statutory calculations (e.g., haircut compounding and capital adequacy deductions) represent the primary bottleneck for small 7B models due to active working memory and token representation limits. As outlined in Section 15, bridging this gap requires parameter scaling (14B/32B/70B) on institutional GPU compute alongside a sandboxed Python REPL tool execution pipeline.

15. Future Scope & Scaling Roadmap

1. Foundation Model Scaling on Institutional Compute (14B–70B): While our 7B model leads all open base models at 80.98%, fine-tuning larger open models like DeepSeek-R1-Distill-14B/32B or Meta-Llama-3.3-70B on institutional GPU compute (e.g., A100/H100 clusters) will expand transformer depth and working memory, projected to achieve >91.5% accuracy and surpass proprietary cloud models on-premises.

2. Program-Aided Compliance (Python REPL Sandbox): To solve complex multi-step capital adequacy and haircut compounding calculations (where 7B models score 69.18%), integrating an automated Python code execution sandbox inside <think> reasoning traces will eliminate arithmetic errors and push numerical accuracy from 69.18% to 98%+.

3. Online Group Relative Policy Optimization (GRPO): Replacing static offline DPO with DeepSeek-style online GRPO using verifiable rule-based reward functions will directly reinforce exact statutory citations and mathematical correctness without reward model collapse.

4. Autonomous 24/7 Regulatory Monitoring Daemon: Deploying background crawlers to monitor daily gazette notifications and circular releases from all 13 authorities, incrementally updating the SQLite WAL index with real-time SSE compliance alerts.

PART K — References

16. References (8 Primary Literature Sources in IEEE Format)

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